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Driving Decarbonization at BMW

BMW Group's (hereafter: BMW) CEO Oliver Zipse looked out his office window from the 22nd floor of the company's iconic four-cylinder-shaped headquarters in Munich, Germany, in June 2022. He was reflecting on his vision for the company's future, which included producing cars with alternative powertrains—diesel and petrol (gasoline) internal combustion engines (ICE), plug-in hybrid electric vehicles (PHEV), and battery electric vehicles (BEV)—so that customers could choose the one that best fit their preferred driving experience, energy infrastructure, range, price, and other criteria.

Meanwhile, automotive investors were increasingly focused on the transition to electric mobility and were scrutinizing legacy automakers' ability to transition from ICE vehicles to electric vehicles (EVs), which included BEVs and PHEVs. Regulators were also exploring ways to reduce carbon emissions associated with transportation, and were increasingly focused on accelerating the transition to EVs through rebates, fleet mandates, and—more recently—by setting a year by which new ICE vehicles would no longer be allowed to be sold. For example, the European Union (EU) was discussing banning the sale of new ICE vehicles by 2035. Several automakers had recently announced commitments to sell only EVs, under certain conditions, by around that year (see **Exhibit 1** for a sample of announcements). In contrast, Zipse was unwilling for the BMW brand to announce such a date, and instead focused on a flexible strategy that offered multiple powertrains for world markets. His decision was motivated by the uncertainty surrounding how quickly and comprehensively EV charging infrastructure would be deployed in a number of major markets, and by his sense that customers would continue to value driving long distances largely uninterrupted, whether for private or professional purposes. The BMW Group had announced that all MINI and Rolls Royce vehicle models would be fully electric starting in 2030, believing that those customers' needs and usage patterns—including their predominant use for shorter trips, mostly in urban areas, and the availability of other cars in the driver's household—would allow for a full switch.

BMW's approach to sustainability focused not just on reducing vehicle tailpipe emissions, but also decarbonizing materials from its supply chain and its production. This meant focusing on recycled and recyclable materials pursuant to the notion of a circular economy, and to shifting auto manufacturing plants' heat and electricity to rely on renewable energy sources. For Zipse, this end-to-end focus was

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more holistic and carefully considered, rather than ignored, tradeoffs between decarbonizing different elements. For example, while EVs avoided tailpipe emissions, their emissions shifted upstream to the charging infrastructure, whose emissions intensity depended on whether electricity was generated by fossil fuels, renewables like wind and solar, or nuclear power. Moreover, the production of EV batteries caused substantial carbon emissions and other environmental concerns associated with mining and processing of lithium and cobalt, key battery ingredients. BMW had introduced the i3 in 2013 as one of the world's first mass-market BEVs, and had since delivered more than 1 million EVs to customers. In 2021, BMW delivered 328,314 EVs, which exceeded the EV deliveries of competitors like Mercedes-Benz and Audi.

Zipse believed that providing multiple powertrain systems afforded BMW critical flexibility, and that focusing on lifecycle carbon emissions would allow BMW to provide customers "with premium individual mobility that includes an outstanding level of sustainability." He also believed this strategy would increase BMW's market share and financial resilience. Yet investors seemed unconvinced, with BMW's market value trading at only four times earnings, and its market-to-book ratio at just 0.7, which meant that investors' market value of BMW equity was less than the company's book value of equity on its balance sheet. Meanwhile, investor enthusiasm was much higher for the upstart California-based automaker Tesla, which produced only BEVs: Tesla's market value exceeded the combined market value of the ten next largest automakers. Moreover, regulators were still focusing only on tailpipe emissions rather than focusing on minimizing lifecycle emissions. Zipse believed an ICE ban could have negative climate impacts due to BEVs' greater carbon emissions upstream in the supply chain and because the lack of charging infrastructure would lead some people to retain their older, less fuel-efficient cars longer. How could he better communicate his flexible powertrain strategy and the emphasis on reducing lifecycle emissions to investors and regulators?

Company Overview

BMW was founded in 1916 in Munich, the largest city in the German state of Bavaria, as an aircraft engine manufacturer. In 1922, the company built a plant in Munich to begin producing automobiles and engines. BMW was an abbreviation of Bayerische Motoren Werke, which translated to the "Bavarian Engine Works Company." The company's slogan "Freude am Fahren" translated to "sheer driving pleasure," which came to be known in the United States as the "ultimate driving machine."[1] BMW emphasized engineering excellence and the customer driving experience, and had developed a strong fan base and loyal customers; 45% of BMW owners chose a BMW again as their next car.[2]

In the 1970s, BMW introduced the 3-, 5-, and 7-series models that became BMW's core heritage and established an image of car luxury and power. In 1972, BMW introduced the M series that featured more acceleration and faster top speeds for "performance obsessed drivers." BMW acquired the Rover Group in 1994 and sold it in 2000, retaining the MINI brand. BMW acquired the Rolls Royce brand in 1998 and started producing Rolls-Royce branded cars in 2003.[3]

Culture BMW's company culture emphasized commitment, engineering, and loyalty. BMW had a "get things done" mindset, and tended to under-promise and over-deliver. A common phrase among BMW managers was, "If we make a promise, we will deliver." BMW emphasized engineering excellence and a spirit of continuous improvement to provide the best driving experience and fuel efficiency. BMW management was a tight-knit group who knew each other well, with most top managers having risen from within the company over a period of decades. Many were German engineers and shared a fondness for quantitative analysis. Thomas Albrecht, Head of Efficient

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Dynamics, reflected, "My colleagues don't mind ambitious targets. But you need to explain to them the rationale. Convince them with measurements and numbers from competitors."

Financial Performance Financially, BMW had gone through ups and downs. In 1959, BMW was nearly bought out by Mercedes-Benz, but instead was rescued by a $1 million investment from the Quandt family. In 2022, the Quandt family owned 42% of the company and was its largest shareholder, followed by traditional investment managers such as Vanguard and Amundi, which in aggregate held 19% of the company's shares. These long-term investors focused more on dividends than short-term variations in share price. Zipse explained, "A high level of profitability and a growing market share are key performance indicators for our business. In the long run they are also the basis for a positive share price development and a high level of dividend payout. This is what most of our investors are seeking." BMW relied on bonds to raise funds, and maintained an "A" S&P credit rating, the best among European automakers.[4]

Sustainability and Initial EVs BMW's focus on sustainability started in the 1970s, as the first automaker to appoint an environmental officer responsible for factory compliance with pollution laws (see **Exhibit 2** for a timeline of key sustainability milestones).[5] In 1972, BMW's first electrically powered test vehicles were driven in the 1972 Munich Olympics to provide support and film the marathon.[6] In 2000, BMW defined sustainability as a guiding principle of corporate strategy, and released its first sustainability value report in 2002.[7] In 2007, BMW introduced the concept of efficient dynamics, which referred to providing the ultimate driving experience with maximum efficiency. This concept committed BMW to seek to consistently reduce fuel or energy consumption while also improving the driving experience.[8]

Also in 2007, BMW internally launched Project i, a program to develop a mega-city BEV that would incorporate sustainability. At that time, the only BEV on the market was the high-end Tesla Roadster sports car, which was not mass-produced. Project i resulted in the launch of the BMW model i3 in 2013. The i3 was a BEV manufactured in a production plant that used renewable energy, and was designed to have the lowest lifecycle carbon footprint of any car sold at that time. The car body was made of lightweight carbon fiber instead of steel, and the car's design featured many examples of circularity – reusing and recycling materials as long as possible. As a result, 95% of the car was recyclable, and its door panels could be composted.

The i3 had a futuristic look that differed from typical BMW models (see **Exhibit 3** for photos), and provoked mixed reactions. A fund manager at Union Investment, a top 20 BMW shareholder, complained that BMW "burnt €2bn on the i3, and the market was not ready yet."[9] Kai Langer, Head of BMWi Design and a founding team member of Project i, recalled that the "i3 was polarizing and disrupting. We wanted it like that. It is a big sound, surprise, it gets people's attention. I was always interested in how you can make big steps. You need to rattle the cage in order to renew heritage. The innovation of today is the heritage of tomorrow."

In 2014, BMW launched the i8, a plug-in hybrid sports car, to demonstrate the possibilities of electromobility technology. Langer elaborated, "It showed the ability of being sustainable and also something that is more traditionally related. It showed you don't have to sacrifice fun." In 2021, BMW introduced the i4, which Langer described as a "more traditional, beautiful car. This is where we hit the core design with technology and ideas that ten years ago were way too progressive."

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BMW in the 2020s

Strategy In the 2020s, the automotive industry was rapidly transforming from traditional ICEs to EVs, with rapid growth in both EV sales and model introductions (see **Exhibit 4** for the market growth of EVs). That said, consumers considering a BEV listed range anxiety and the lack of public charging infrastructure as their biggest concerns.[10] As of 2021, charging station availability varied greatly across the EU, from the Netherlands' densest network of 47.5 chargers per 100 km of roads to Lithuania's least dense featuring only 0.2 per 100 km (see **Exhibit 5** for charging station density in EU countries).[11] In 2021, recharging an EV battery to 80 percent took 30 minutes on average, whereas refilling a gas tank took less than five.[12]

BMW's strategy focused on flexibility and circularity. Their flexible powertrain strategy meant that BMW models were being designed with various alternative powertrains to suit different customers' preferences. For example, urban dwellers with shorter commutes and better access to charging stations might prefer BEVs, whereas those commuting longer distances or living in rural areas with less access to charging stations might benefit from ICEs (see **Exhibit 6** for average daily miles driven). In 2020, BMW's X3 compact sport utility vehicle (SUV) was the first model available where customers faced the options across powertrains: the X3 was available with an ICE (diesel or petrol), as a PHEV, or as a BEV called the iX3.

While BMW maintained a flexible powertrain strategy for the BMW brand, it made a commitment that all sales in the MINI and Rolls Royce brands would be fully-electric by 2030. Thomas Becker, Vice President of Sustainability and Mobility, explained, "The MINI typically serves customers who are urban, use it as a second vehicle, don't use it for long trips, store it in their home garage, and are predominantly located in OECD markets – all factors that support the availability of EV infrastructure and avoid problems of battery range. The Rolls Royce is a brand where you don't expect an engine roar (like BMWs), where customers aren't price sensitive, and for longer distances will use other transportation modes rather than their Rolls Royce."

BMW's flexible powertrain strategy and its decision to not set a date by which it would phase out ICEs was somewhat reflective of the company's experience with other technology transformations facing the automotive industry. BMW had jointly invested € 1 billion with Daimler in 2018 in the FREE NOW ride-sharing service's parent company YOUR NOW,[13] but recognized impairment losses of € 240 million in 2019 as the outlook soured.[14] Zipse reflected on the impairment, "Some years back it seemed to be common sense amongst industry observers that ride sharing and autonomous driving technology would make traditional automakers irrelevant. But the reality is that several years later, most ride sharing companies are struggling to make a profit. And while automated driving functions are an essential technology that we also offer to our customers, fully autonomous driving is still far away from implementation. You need to be mindful about the pace of change and allow yourself the flexibility to be competitive no matter where the market goes. The same is the case with powertrain technology."

BMW's strategy also emphasized material circularity and measuring and reducing carbon emissions throughout the life cycle of its vehicles, including the impact of their suppliers' producing materials and components, of BMW's own production processes, and of their customers' use of BMW vehicles. Becker commented, "We try to anticipate what will be next, and by 2030, our customers will stop comparing electric cars to conventional vehicles. They will want to know the difference between electric car A and electric car B. Just having zero tailpipe emissions will not be enough, and so we need to think about efficiency and upstream emissions."

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Major Markets In fiscal 2021, BMW's revenue totaled about € 111 billion (see **Exhibit 7** for financial statements) and its largest market segments were Europe, the US, and China (see **Exhibit 8** for major market segments). The fastest growing market was China, where sales had increased by 8.9% between 2020 and 2021, providing 33.6% of BMW's total revenue. BMW had been expanding its presence in China through two joint ventures that produced BMW engines and automobiles.

Regulation Transportation was one of the largest sources of carbon emissions, resulting in nearly 16% of total global emissions in 2018, 75% of which was from road transportation.[15] In response to the Paris Climate Accord goal to keep global average temperatures from exceeding a 2 degree Celsius increase by the end of the 21st Century relative to preindustrial levels, many countries had set targets to shift the automobile market toward EVs. Within the U.S., California passed a rule in August 2022 that required all new vehicles sold by 2035 to be Zero-Emission Vehicles (e.g., BEV, fuel cell electric). More than a dozen other states, including New York and Massachusetts, were considering following California's rule.[16] U.S. President Joe Biden announced in 2021 a nationwide target that 50% of vehicles sold in 2030 in the U.S. would be BEVs, PHEVs, or fuel cell electric vehicles.[17] In China, the State Council's vehicle development plan for 2021-2035 included a target of 20% EV by 2025.[18] In Europe, as part of its "Fit for 55" plan to reduce emissions by 55% by 2030 and achieve climate neutrality by 2050, [19] the EU set a target to reduce vehicle tailpipe CO_2 emissions by 55% by 2030, and in June 2022 proposed to eliminate by 2035 the sale of new vehicles with tailpipe emissions.[20] If the European Council and Parliament agreed, this would essentially ban the sale of new ICE vehicles and PHEVs starting in 2035.

The introduction of carbon pricing regulations would also have an impact on the automotive industry. An increasing percentage of greenhouse gas (GHG) emissions around the world were being regulated and priced through carbon taxes or cap and trade systems, with the percentage increasing from less than 5% to more than 21% in the first two decades of the 21st Century.[21] Since 2005, the EU maintained an Emissions Trading System (ETS) for energy-intensive industry sectors that included many automotive suppliers, such as steelmakers. The EU ETS was a cap and trade system consisting of an EU-wide emissions cap (maximum allowed emissions), and each firm received an allocation of emissions credits. Firms could trade these credits, selling them when their allocation exceeded their emissions, and purchasing them when their emissions exceeded their allocation. Additionally, the EU "Fit for 55" plan also included a Carbon Border Adjustment Mechanism (CBAM), which meant that importers of carbon-intensive materials and products produced outside of the EU would be required to pay a tax at the border to adjust for the differences in GHG emissions tax for the goods produced within and outside of the EU. The application of this concept to automotive supplies with embedded emissions from such materials in the future was highly likely. Florian Weig, then Senior Vice President Corporate Strategy, reflected on that trend, "I asked my colleagues: Would you want to incur a 100 euros per ton of carbon when we sell a 3-series or would you prefer that we reduce our carbon emissions? There was unanimity among the management team that decarbonization was the way to go."

Competitive Landscape Historically, BMW's main competitors in the luxury car market were Mercedes-Benz and Audi. Tesla's entry transformed the market by bringing technical feasibility to EVs. The cost of lithium-ion batteries declined by 89% over the decade leading to 2020, and EV sales reached 6.7 million by 2021, which was nearly 9% of global car sales.[22,23] A growing number of legacy automakers were starting to offer EVs. While BMW was designing vehicle models like the X3 that could accommodate various powertrains, competitors like Mercedes-Benz and Audi were introducing EVs with distinct designs, such as the EV-only Mercedes EQ series.[24]

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In 2021, Tesla had the largest market share in the EV market, at 13.8%, and BMW ranked seventh at 4.8% (see **Exhibit 9** for EV market share). Tesla's share price had appreciated much more than legacy automakers. As of the end of 2021, Tesla's market value was 36 times its book value, yet its free cash flow was only 20% of its market value, and it paid no dividend. In contrast, BMW and Mercedes-Benz had low market-to-book ratios of around 1, but had higher free cash flow and paid dividends (see **Exhibit 10** for comparative ratios and metrics).

In response to the growing regulation on the conversion to EVs, many automotive brands had released an "end date" to phase out ICEs in specific markets or product segments: Mercedes-Benz, Bentley, Buick, Fiat, Volvo all set a 2030 deadline, and Audi, 2033.[25] One of the largest automakers, General Motors announced that it "aspires to eliminate tailpipe emissions from new light-duty vehicles by 2035."[26] Additionally, during the UN Climate Conference of Parties "COP26" summit in 2021, several governments and companies, including Daimler, Ford, General Motors, and Volvo Cars, signed a declaration to "work towards all sales of new cars and vans being zero emission globally by 2040, and by no later than 2035 in leading markets."[27] Other companies, including BMW, Toyota, and Volkswagen, did not join the pledge.

Zipse described what he thought the situation would be in 2035 if there were a ban on ICE, "There would be a number of people who won't be able to charge because of a lack of infrastructure, another group of 'combustion guys' would simply continue to prefer ICE vehicles, and others couldn't be served with BEV vehicles because of a limited supply of battery materials. As a consequence, these customers would continue to drive their old cars - and leaving old cars longer in the market can have negative climate effects – the contrary of what we want to achieve." Facing these constraints, Zipse believed that instead of an ICE ban, "Reducing CO_2 emissions throughout all technologies year by year is of fundamental importance." Moreover, BMW's committing to a voluntary end date for ICEs would conflict with its strategy of technology openness, and would leave the ICE market to other companies who did not agree to phase them out.

BMW's decision to not commit to phasing out ICEs by a given year evoked skepticism in the public and the investment community. An August 2021 *New York Times* article commented, "The perception that BMW is an electric vehicle laggard helps explain why investors have begun to sour on the company's shares, which fell even after the company reported a healthy quarterly net profit this month of 4.8 billion euros, or $5.7 billion. BMW shares have tumbled 18 percent since early June."[28] In May 2022, impact investor Inyova wrote that BMW ran "a strong risk of becoming the Blackberry of the automotive industry," referring to the cellphone manufacturer that rose to prominence in the 1990s and early 2000s but then lost almost all market share, and that Inyova would "keep monitoring BMW closely. If we don't see it embracing a low-carbon strategy, it may be removed from the Inyova universe."[29]

Veronika Rösler, Head of Investor Relations, described these comments as a perception gap, "I would say our approach and other companies are very similar, starting from multi powertrain architectures and moving towards purely electric architectures. However, it is different in terms of perception, communication, and how that is reflected in capital market views." She added, "It reflects an element of the BMW genes, that we want to get things done, and then we talk about it. If I talk about something, I have to be 100% sure that I am going to achieve that exactly in that way. And so this is something which I think probably is connected to the way we communicate and also then the way things are perceived."

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Decarbonization at BMW

BMW's decarbonization efforts focused on reducing GHG emissions throughout the vehicle's life cycle: materials and components from suppliers, BMW's own manufacturing and assembly processes, and vehicles' usage and end-of-life disposition. Becker emphasized, "We must always ask ourselves: How sustainable are we? And we must be able to answer this question and provide proof at every single point, authentically, transparently and measurably, including to third parties."

BMW measured its carbon footprint using the Greenhouse Gas Protocol (GHG Protocol), which was the world's most widely-used GHG accounting standard. The GHG Protocol categorized emissions into three scopes. Scope 1 referred to emissions resulting "from sources that are owned or controlled by the reporting entity," which for BMW included the combustion of natural gas used to generate ambient heat and process heat, such as in their paint shops.[30] Scope 2 referred to emissions from "activities of the reporting entity but occur at sources owned or controlled by another entity," which were associated with the purchase of electricity, steam, heat, or cooling, and for BMW included the electricity purchased to supply its office buildings and production plants. Scope 3 referred to other indirect emissions not included in scopes 1 and 2 that were associated with the company, including upstream emissions associated with its suppliers' producing and shipping components and other materials, and downstream emissions associated with the fueling or charging of BMW vehicles, and vehicle disposition at their end-of-life (e.g., recycling, landfilling). While most companies measuring their carbon footprint included scopes 1 and 2, some omitted scope 3 entirely while others chose to focus on select scope 3 items.

Exhibit 11 reported BMW's GHG emissions for each of the three scopes. Scopes 1 and 2 combined represented less than 1% of the company's total emissions of 122.5 million metric tons of CO_2e in 2021.[a] The vast majority of the emissions associated with BMW vehicles were within scope 3, with 15% of total emissions coming from the supply chain, 81% from the vehicle's use, and 3% from other scope 3 emissions. These emissions figures were reported in BMW's 2021 Integrated report, for which BMW's financial auditor PricewaterhouseCoopers GmbH provided a limited assurance over the non-financial information.[b]

In 2020, BMW set new targets for 2030 that were validated by the Science-based Targets Initiative (SBTi) (see **Exhibit 12** for targets).[c] BMW chose a time horizon for 10 years, "which we believe can gain credibility by laying out a path and year-over-year show improvement. Moreover, we thought it would be more credible to set a 10-year target than what most of our competitors do in 15- or 20-year targets as we would still be around to see the completion of the targets," Weig explained. BMW further joined the Business Ambition for 1.5°C campaign by committing to the SBTi Net Zero Standard, which

[a] CO_2e refers to carbon dioxide equivalent. In addition to carbon dioxide, emission figures included impacts from other greenhouse gases, such as methane and nitrous oxide. The global warming impacts of the other greenhouse gasses were converted into the equivalent amount of carbon dioxide with the same global warming potential. The conversion ratios were based on the global warming potential (GWP) values released by the UN Intergovernmental Panel on Climate Change (IPCC).

[b] Limited assurance provided opinion in a negative form of expression that nothing came to the auditor's attention to indicate that the information disclosed was materially misstated. Relative to limited assurance, reasonable assurance was a higher level of assurance that provided a positive form of expression that the information was presented fairly.

[c] SBTi was an increasingly widely-adopted standard for setting emissions targets that aligned with a 1.5 degree scenario as part of the Paris Agreement goals. Companies calculate science-based targets based on either sector-specific decarbonization pathways or on a percentage reduction in absolute emissions. The targets should cover a 5 to 10 years horizon from the date the target is publicly announced. SBTi verifies company targets and provides certificates for companies that meet the standards.

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involved reducing absolute emissions by at least 90% compared to the base year and attaining net zero emissions by latest 2050.

To achieve these emissions targets, BMW created sub-targets. As Roberto Rossetti, Head of Vehicle Life Cycle, explained, "We break it down by vehicle and by components, and for our engineers who design one component, each of them knows their individual targets. The same with the purchase team that works with all the suppliers." These sub-divisions then explored and pursued decarbonization opportunities to reach their targets.

Scopes 1 and 2

BMW's total scope 1 and 2 emissions for 2021 were 699,713 and 134,849 metric tons of CO_2e, respectively. These emissions were relatively straightforward to calculate, as they were based on invoices for fuels consumed (for scope 1) and electricity purchased (for scope 2), and had been measured for over 20 years with relatively well-established processes. For each of their production plants, BMW knew all sources of emissions, such as the amount of electricity used and the cubic meters of natural gas that went into a process. For its other facilities, such as sales centers in different regions, BMW obtained emissions associated with their electricity consumption directly from the utility, or when unavailable, converted its electricity usage to emissions using national averages of the emissions intensity of electricity.

BMW had set an ambitious goal to reduce by 2030 scopes 1 and 2 emissions per vehicle by 80% below 2019 levels. Becker commented, "We are more aggressive with our own emissions than the goals we are setting for the suppliers. This is *our* CO_2, and we want to bring it down to be authentic when we talk to our suppliers and require them, for example, to abandon natural gas."

To gauge what was needed to achieve this target, BMW constructed a carbon abatement cost curve (see **Exhibit 13** for generic examples for ICE vehicles and BEVs). The carbon abatement cost curve was a tool to illustrate the cost and carbon abatement potential of various decarbonization activities. The horizontal axis showed each opportunity's abatement potential, measured in metric tons of CO_2e. The vertical axis showed the average cost per metric ton of abated carbon. Such charts typically displayed the decarbonization activities from the lowest to the highest average cost per metric ton of abated carbon. Some activities might have a negative cost, meaning they were win-wins: they saved money and abated carbon.

For BMW's scopes 1 and 2 emissions, some opportunities that had negative costs were energy efficiency projects, such as some low-cost process improvement activities that made the manufacturing plants more energy efficient and offered rapid payback periods via the energy cost savings they yielded. BMW's opportunities with close-to-neutral costs included purchasing green electricity by signing renewable energy power purchase agreements (PPAs) at a relatively low cost premium. More costly opportunities were projects that addressed harder-to-abate problems, such as efforts to generate less carbon-intensive process heat in production plants, such as via electrification or the use of "green hydrogen"—hydrogen produced via electrolysis powered by renewable energy.

BMW's staff worked with external consultancies and experts to develop an initial abatement cost curve in 2020 and estimated that an 80% reduction in scopes 1 and 2 emissions was economically feasible. Yet after in-depth analysis and more precise information, achieving these objectives turned out to be more challenging than was initially anticipated. One reason was that the decarbonization opportunities were less available than initially anticipated, and that the feasible set of decarbonization projects varied across BMW's production plants. For example, the availability and cost of renewable electricity depended on the weather conditions in the region that affected solar and wind power.

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Additionally, electrifying to replace fossil fuels could be constrained by their production plants' infrastructure. For example, most production plants traditionally relied on natural gas to generate the vast amount of heat required in the paint shop to dry vehicles after the paint was applied, and retrofitting to electrify this process was expensive and could require shutting down the plant for several months. In contrast, BMW's newly-built plant in Debrecen, Hungary, was designed to be a carbon-free factory from the beginning. Its paint shop was designed to be fully electric, and the plant purchased electricity generated only from renewable sources.

Another issue with such initial abatement cost curves was that they did not reveal potential interdependencies across projects. As Michael Bengsch, responsible for Corporate Strategy - Sustainability and Mobility, explained, "For example, to abate the emissions from the use of natural gas, assuming all three are available, I could buy biomass certificates to make it CO_2 neutral, I could replace the natural gas with hydrogen, or I could switch to electric heat and use green power. All three show up in this curve with different costs, but if you pull one lever, you will exclude the other."

Ingo Janka, Head of Strategy, Planning, Environmental Protection, described, "Because BMW is systematically investing in energy efficiency, promoting the substitution of natural gas by renewable energy sources, and taking advantage of the possibilities opened up by digitalization, we are going to achieve our Scope 1 and 2 reduction targets. This is what BMW really stands for. This is why I am very proud to work for this company." Weig added, "The one amazing thing to see was the amount of unity and determination of the leadership team behind these efforts. I never felt that the executive team wavered for a minute."

Scope 3 Upstream: Supply Chain

BMW's scope 3 upstream (supply chain) emissions for 2021 were 18.3 million metric tons of CO_2e. Most of this was from the production of the steel, aluminum, plastic, and batteries required to produce their vehicles. Measuring these emissions was an ongoing learning process. BMW maintained records of all materials purchased from suppliers, and in most cases estimated their associated emissions using industry average emission factors from GaBi. GaBi was a software application that contained life cycle assessment[d] data for more than 15,000 processes based on primary industry data, and was used by more than 10,000 organizations.[31] GaBi provided average emissions factors per unit of a given material produced with a specific technology in a particular region. For example, the emissions factor it provided per ton of steel production depended on the production technology (e.g., warm bending or cold bending) and the region in which it was manufactured (e.g., China or Europe).

When available, BMW used primary data on carbon emissions that was provided directly by the supplier. Yet few suppliers had obtained emissions data from their own suppliers, especially those four to five tiers down the supply chain. Given this challenge, less than 10% of suppliers provided primary data to BMW. For the remaining suppliers, BMW would calculate emissions reductions from specific decarbonization actions taken by the supplier, and deduct that from the average emissions amount calculated using the GaBi database. Going forward, the goal was to transition to increasingly use primary data from suppliers. To facilitate this, BMW became a founding member of a data-sharing initiative called Catena-X,[32] an open data ecosystem for the automotive industry supported by the German government, where companies throughout the value chain could exchange carbon emissions information using a standard approach.

[d] Life cycle assessment is a methodology to measure the environmental impacts associated with a product throughout its lifespan, including extracting and processing raw materials, manufacturing, distributing, using, maintaining, and disposing or recycling.

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BMW's target for scope 3 upstream emissions was a 22% reduction per vehicle by 2030 relative to a 2019 baseline. Patrick Hudde, Head of Sustainability Supply Chain, Raw Material Management, Energy, noted that this would be especially challenging to achieve, "In order to achieve the reduction target of 22%, the reduction targets for the average vehicle project after 2025 will need to be much higher (up to 60%-70%) because the average unabated CO_2 footprints of those vehicle projects are much higher (up to 40%-50%) due to the higher share of BEVs in the product portfolio". For the average BEV at BMW, the battery accounted for 26% of its supply chain carbon emissions (see **Exhibit 14a** for emissions by material). The emissions impact of batteries varied by battery size and where they were produced, since nearly half of batteries' lifecycle emissions came from the electricity used in their manufacturing and assembly.[33] As a result, Weig observed, "The BEV upstream carbon is about 50% higher than that of the ICE. The 3 series might have less than 10 metric tons of CO_2e per vehicle while the i4, after some decarbonization measures and depending on the size of the battery, has closer to 15 metric tons of CO_2e."

Weig described BMW's two stage strategy to reduce upstream carbon emissions, "First stage, drive all suppliers towards renewable energy. This makes economic sense. Second stage, move suppliers from primary materials to recycled materials. How? Make the use of recycled materials the default choice, and make them explain why they need to use primary. Why should we do this even though it might add some cost? Because circularity is the biggest insurance against raw material scarcity and geopolitical risk, and we believe more economic in the long term."

To achieve these targets, BMW worked closely with its suppliers. When sourcing components, BMW provided suppliers with targets for cost, quality, and carbon reduction. BMW also provided guidance on reaching the carbon reduction target, such as by using green energy in production, or using secondary materials. BMW had signed agreements with suppliers to implement 30 million metric tons of carbon-reduction initiatives for supplies that would be purchased before 2030—most of which was from shifting to renewable energy, particularly to produce batteries and aluminum.

BMW also engaged in various due diligence and verification activities to make sure their suppliers' promised carbon reductions were credible. For example, BMW engaged third-party auditors to spot check its suppliers' emissions claims. Financial reporting auditors also validated that the emissions savings reported on annual reports were based on reliable contracts with suppliers. A key challenge was the lack of standard approach to measure emissions in supply chains. Becker commented, "It's really a work under development. There are no standards, no state-of-the-art. So that's why we are working on that with Catena-X together with many partners, and we want to engage policymakers to make sure that they accept the outcome of this."

Without uniform standards, BMW needed to make decisions about whether certain carbon reduction approaches should count toward their emissions targets. One example was the use of mass balancing, where a supplier could allocate all carbon abatement to a few units of output and declare the reduction achieved. For instance, after one steel company increased the efficiency of their blast furnace to reduce emissions by 10%, it then sought to sell 10% of its steel as carbon-free at a higher price. Should BMW purchase and consider that steel to be carbon-free?

BMW created a separate abatement cost curve for scope 3 upstream emissions to compare opportunities to decarbonize its supply chain. Decarbonization projects with lower costs included the use of recycled materials and the use of renewable energy in the supplier's production process. The use of renewable energy had the largest abatement potential of around 50% of the carbon emissions, and given the falling cost of renewables in many areas where BMW suppliers were located, was assumed to have a close to zero abatement cost. The higher abatement cost options included technologies that sought to transform hard-to-abate sectors, such as producing less carbon-intensive "green" steel.

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Despite its higher abatement cost and higher price—green steel sold at a 20 to 30% price premium in 2021—BMW was planning to procure some green steel because its development was strategically important.[34] Rossetti explained, "It gets the supply chain going. It sends a signal to the steel industry: we need green steel." BMW was also working with a supplier that was using hydrogen technology to decarbonize the steel-making process, and BMW expected to begin receiving green steel produced with this technology in 2025.

Using recycled materials often provided a relatively low-cost way to decarbonize the supply chain. In 2022, nearly 30% of BMW's vehicle components had recycled content. Recycled aluminum had the highest ratio, with over 50% of aluminum sourced from secondary materials. Recycled aluminum was particularly beneficial from a carbon emissions perspective because primary aluminum produced with fossil fuel based electricity could result in the emissions of 20 tons of carbon per ton of aluminum, a figure that declined to 4 tons if renewable energy was used, and to 0.5 tons if recycled aluminum was used.[35] However, BMW and its suppliers faced challenges recovering end-of-use materials to recycle. For example, finding more post-use aluminum to produce higher percentages of recycled aluminum was challenging. Not only did automotive suppliers face competition to source aluminum from scrapped vehicles from other industries such as beverage can manufacturing, but automotive components required high-grade aluminum that was undiluted and uncontaminated—conditions not required by other industrial users of recycled aluminum. As such, BMW and other automakers felt it was important to develop closed-loop systems to enable vehicles' materials to be returned to the automotive supplier chain.

Batteries were another example where recovering recycled material could play a big role in BMW's decarbonization efforts. The procurement of batteries was a highly competitive market, with most large battery manufacturers located in Asia. Stefan Brandlhuber, Head of Purchasing, QMP Cells, Modules, Storage Systems, reflected on the importance of battery procurement, "We are talking about partners, not suppliers, because this is a long-term relationship. We and our partners have a global footprint and we need to innovate to produce batteries that are safe, efficient, and with low carbon footprint. Given market demand you need to convince the battery manufacturers why they should work with us. We say we want to build the greenest BEV and that excites them. They want to be pushed technologically." Brandlhuber's team had 120 people across the globe and he was focusing on hiring more people in a market that exhibited "a war for talent."

The production of batteries required several rare minerals including cobalt, lithium, and nickel (see **Exhibit 14b** for minerals used in vehicle production). Their availability was a constraining factor in mass production of EVs, they created significant risks given their price volatility and concerns with human rights and labor practices associated with their mining and processing. Mining production and processing of these minerals was geographically concentrated. For example, the Democratic Republic of the Congo (DRC) produced 70% of the world's cobalt, and China processed 50 to 70% of the world's lithium and cobalt.[36] The DRC's cobalt mines elicited concerns about worker safety and mining pollution threatened the health of neighboring communities.[37,38] BMW had a strategy of sourcing raw materials directly and had entered into long-term mining contracts for lithium and cobalt. Brandlhuber explained, "BMW is focusing on the complete value stream of battery cells to secure capacity of critical minerals sourced from mines that are certified as meeting environmental and human rights standards."

Recycling batteries was strategically important to BMW because using secondary materials in battery production reduced the reliance on rare virgin minerals. The challenge was in finding and retrieving used batteries, especially given other automakers were also seeking to do so. Further, since EVs were relatively new, large quantities of expired EV batteries were not expected to be available to recycle for many years. In China, regulations held automakers responsible for the recovery of recycled

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batteries, which ensured BMW could collect old batteries from BMW vehicles and use them in new production. The EU and U.S. did not have similar policies in place. In addition to recycling, BMW was also conducting internal research and development on battery production, and was developing technologies to avoid the use of rare minerals like cobalt.

Scope 3 Downstream: Use-phase

Scope 3 downstream emissions associated with a vehicle's use-phase represented the vehicle's largest source of emissions. The use-phase emissions associated with a vehicle's lifespan were typically calculated assuming vehicles would be driven 200,000 km (10 years for 20,000 km per year). Use-phase emissions stemmed from fuel combustion in ICE vehicles, from upstream emissions of electricity generation that charged BEVs, and from a combination of these factors for PHEVs. BMW measured use-phase emissions from driving tests and calculated a fleet wide average that was weighted based on the volume of its sales in the EU, U.S., and China, which represented over 80% of BMW's sales in 2021. Driving tests to estimate energy usage were conducted based on standards such as the Worldwide Harmonised Light Vehicle Test Procedure (WLTP) used in China and the EU, and the U.S. Environmental Protection Agency 5-cycle test used in the U.S.[39] To estimate emissions associated with BEV charging, BMW used the International Energy Agency's emissions predictions for each country's electricity mix (see **Exhibit 15** for the emissions of electricity in BMW's major markets). The resulting emissions were then increased by 10%, as required by the SBTi to account for potential discrepancies from real emissions. The resulting anticipated lifetime use-phase of BMW vehicles sold in 2021 was calculated to be 100 million metric tons of CO_2e.

BMW had set two targets to reduce scope 3 downstream emissions. The first was to have 50% of its vehicles sold in 2030 be BEVs, with an interim target of 30% by 2025. The second was to reduce its vehicles' use-phase CO_2 emissions by 50% by 2030, compared to a 2019 baseline. Progress toward meeting these targets was driven by strategic decisions that shaped the future sales mix of different powertrains, and by investments in efficiency improvements of the vehicles.

BMW had taken several steps to produce more EVs. BMW's plan was to introduce a variety of electrified models in the coming years, including BEV versions of the BMW 7 Series, the BMW X1, the high-volume 5 Series, the MINI Countryman, and the Rolls-Royce Spectre.[40] BMW also continuously invested in PHEVs, where the BMW X5 xDrive45e introduced in 2019 had an electric range of 54 miles with a battery that is a quarter of the size of a battery for an average SUV BEV.[41] A major design transition was expected to roll out in 2025 with the introduction of a fully electric platform—including powertrain, chassis, wheelbase etc. — called the Neue Klasse. The Neue Klasse would involve a new vehicle architecture geared exclusively towards BEVs, which allowed better customization to fit the BEV powertrain system. It would incorporate the concept of circularity and sustainability by using up to 50% recycled materials, and would be produced in BMW's new emissions-free production plant in Debrecen, Hungary.

As another lever to reduce use-phase emissions, BMW's Efficient Dynamics group was in charge of optimizing the efficiency of the vehicles. "Efficient" referred to how much fuel or energy was needed to propel the vehicle a certain distance. In 2021, the average BMW EV had an electricity power consumption of around 20 kWh per 100 km.[42] Albrecht emphasized, "Every watt counts," when it came to increasing efficiencies for EVs because this could also extend the driving range of EVs. Albrecht explained, "A BMW BEV might be assigned a range of 300 miles but when you go on Edmunds the range in real driving conditions might be 50 miles higher. Some of our competitors might give a higher range but the range is lower in real driving conditions." (See **Exhibit 16** for Edmunds' driving test results). The Efficient Dynamics group was charged with optimizing all aspects of the car to operate as

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efficiently as possible, even under extreme heat or cold conditions. At the same time, efficient dynamics meant more than efficiency: the "dynamics is what you feel when you accelerate the car," Albrecht explained, "Both of these are what we are optimizing."

Efforts to promote efficient dynamics were dedicated to mainly five areas: powertrains, aerodynamics, lightweighting, thermal management, and tires. The largest gains—and thus carbon reductions— had stemmed from improved efficiencies in powertrains, followed by aerodynamics. For example, in 2025, BMW was introducing the 6th generation EV powertrain system that would feature in the Neue Klasse, and the target was to be 30% more efficient than the 5th generation version currently in use. Seventy percent of the Efficient Dynamics team was working on EVs in 2022, but Albrecht believed that within two to three years, the Efficient Dynamics group would focus solely on EVs. He elaborated, "It is always a prioritization. That is what you do as a manager. EV is the future and we need to concentrate on that."

Decisions

Zipse was proud of the decarbonization efforts of BMW. He felt that BMW had a solid plan to make its vehicles the most efficient, dynamic, and sustainable in their class, and that the firm was well positioned to lead in the EV market. Yet some investors remained skeptical and were still using the year automakers were committing to phase out ICEs as an indication of their strategic wisdom. Such concerns were expected to grow as the EU moved to ban the sale of ICE vehicles. Zipse was convinced that analysts were misunderstanding BMW's positioning, noting, "We have been very active in recent years building the capabilities inside the organization to launch EVs in all market segments of our BMW, MINI and Rolls-Royce brands. While this phase was rarely visible for the public, you can now see the result of this strategy: a steep ramp-up of our EV offering that brings us into pole position compared with our direct peers." An analyst report by Credit Suisse confirmed that although BMW was thought as an EV laggard by many investors, it had a higher percentage of sales coming from EVs than its competitor Mercedes-Benz. How could BMW better communicate its strategy and rationale?

Zipse believed BMW's strategy emphasizing flexible powertrains and focusing on lifecycle emissions and circularity would increase its market share and reduce its financial risk. The growth of EVs was increasing automakers' dependence on batteries' raw materials, which sparked financial risks given their limited supply and reputation risks given their concentrated availability from countries with significant social and labor risks. BMW's focus on circularity sought to reduce its reliance on primary rare minerals and bolster its resilience against the volatile battery and battery ingredient markets. Given BMW's market valuation, how could Zipse communicate to investors the firm's growth potentials and financial resilience?

Moreover, BMW's assessment convinced Zipse that the idea of Europe's transitioning all vehicles to fully electric by 2035 would not be as positive for the environment as regulators expected. While BMW and other automakers were working with their suppliers to develop other technologies to reduce supply chain emissions to counteract the battery's impact, they were not there yet—and it might take decades to get there. Moreover, natural gas shortages, due to the war between Russia and Ukraine, had led Germany and other European countries to reopen coal plants that threatened to dramatically increase the carbon intensity of grid electricity. How could BMW convince regulators to increase their attention on carbon emissions beyond the use-phase to also include supply chain emissions, which Zipse felt would help them realize that an ICE ban by 2035 would yield much less economy-wide decarbonization than they thought—and actually result in more carbon emissions in some circumstances?

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Exhibit 1 ICE Phase-out Announcements of Select Automotive Brands

Brand	By	Announcement
Bentley	2030	"Every car we create from 2030 onwards will be a battery electric vehicle."
Buick	2030	"The Buick brand is committed to an all-electric future by the end of this decade."
Ford	2030	"100% of passenger vehicles in Europe will be zero-emissions capable, all-electric or plug-in hybrid by 2026, and completely all-electric by 2030."
Mercedes-Benz	2030	"Mercedes-Benz will be ready to go all electric at the end of the decade, where market conditions allow."
MINI, Rolls-Royce	2030	"MINI and Rolls-Royce will be exclusively all-electric from the early 2030s onwards."
Volvo	2030	"Volvo Cars is committed to becoming a leader in the fast-growing premium electric car market and plans to become a fully electric car company by 2030."
Audi	2033	"As early as 2026, all new models from the Four Rings on the global market will be fully electric. And in 2033, the Four Rings will shut down the production of vehicles with internal combustion engines. One exception could be China, where the company is investigating the possibility of longer production depending on local demand."

Source: Bentley, "Bentley Brand Manifesto," https://www.bentleymotors.com/en/world-of-bentley/beyond-100/bentley-brand-manifesto.html; Buick, "Buick commits to all-electric portfolio by end of decade," https://media.buick.com/media/us/en/buick/news.detail.html/content/Pages/news/us/en/2022/jun/0601-buick.html; Ford Integrated Sustainability and Financial Report 2022, https://corporate.ford.com/content/dam/corporate/us/en-us/documents/reports/integrated-sustainability-and-financial-report-2022.pdf; Mercedes-Benz, "Mercedes-Benz prepares to go all-electric," https://group-media.mercedes-benz.com/marsMediaSite/en/instance/ko/Mercedes-Benz-prepares-to-go-all-electric.xhtml?oid=50834319; BMW Group Report 2021, https://www.bmwgroup.com/content/dam/grpw/websites/bmwgroup_com/ir/downloads/en/2022/bericht/BMW-Group-Report-2021-en.pdf, Volvo, "Volvo Cars to be fully electric by 2030," https://www.media.volvocars.com/global/en-gb/media/pressreleases/277409/volvo-cars-to-be-fully-electric-by-2030; Audi Report 2021, https://www.audi.com/content/dam/gbp2/en/company/investor-relations/reports-and-key-figures/annual-reports/audi-report-2021.pdf.

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Exhibit 2 BMW Sustainability Timeline

Year	Event
1972	For the Summer Olympics in Munich, BMW converted a BMW 1602 into an electrically powered vehicle to chaperone the long-distance walkers and marathon runners as a support vehicle.
1973	BMW became the first automaker to appoint an environmental officer, a person to lay the foundations of preventive environmental protection.
2000	Sustainability became a guiding principle of the BMW Group's corporate strategy.
2002	BMW issued the first sustainable value report.
2009	Sustainability became a corporate target.
2011	Sustainability became an integral part of BMW's procurement process and an essential purchasing criterion.
2012	BMW set ten sustainability goals for the end of 2020.
2013	Introduction of the BMW i3, a resource-saving electric car for urban traffic.
2020	BMW made sustainability central to the strategic direction of the BMW Group.
2020	BMW set ambitious CO2 reduction targets across the entire value chain by 2030.
2021	BMW issued the first integrated report.
2030	CO_2 reduction by at least 40% across the value chain.
2050	Climate-neutral business model across the entire value chain.

Source: The BMW's museum RE:IMAGINE exhibition; BMW Group, *Investor Presentation*, August 2022, https://www.bmwgroup.com/content/dam/grpw/websites/bmwgroup_com/ir/downloads/en/2022/investor-presentation/BMW_Investor_Presentation_2022.pdf, accessed August 2022.

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Exhibit 3 Examples of BMW Electric Vehicle Models

BMW i3 Battery Electric Vehicle



Introduced: 2013
Price in 2013: € 34,950
Horsepower: 170
Acceleration (0-60 mph): 7.3 seconds
Range: 81 miles | 130 km
Mileage: 124 MPGe

BMW i8 Plug-in Hybrid Electric Vehicle



Introduced: 2014
Price in 2014: € 126,000
Horsepower: 369
Acceleration (0-60 mph): 4.2 seconds
Range: 330 miles | 531 km
(electric range: 15 miles | 24 km)
Mileage: 76 MPGe

BMW i4 Battery Electric Vehicle



Introduced: 2021
Price in 2021: € 59,200
Horsepower: 340
Acceleration (0-60 mph): 5.7 seconds
Range: 301 miles | 484 km
Mileage: 109 MPGe

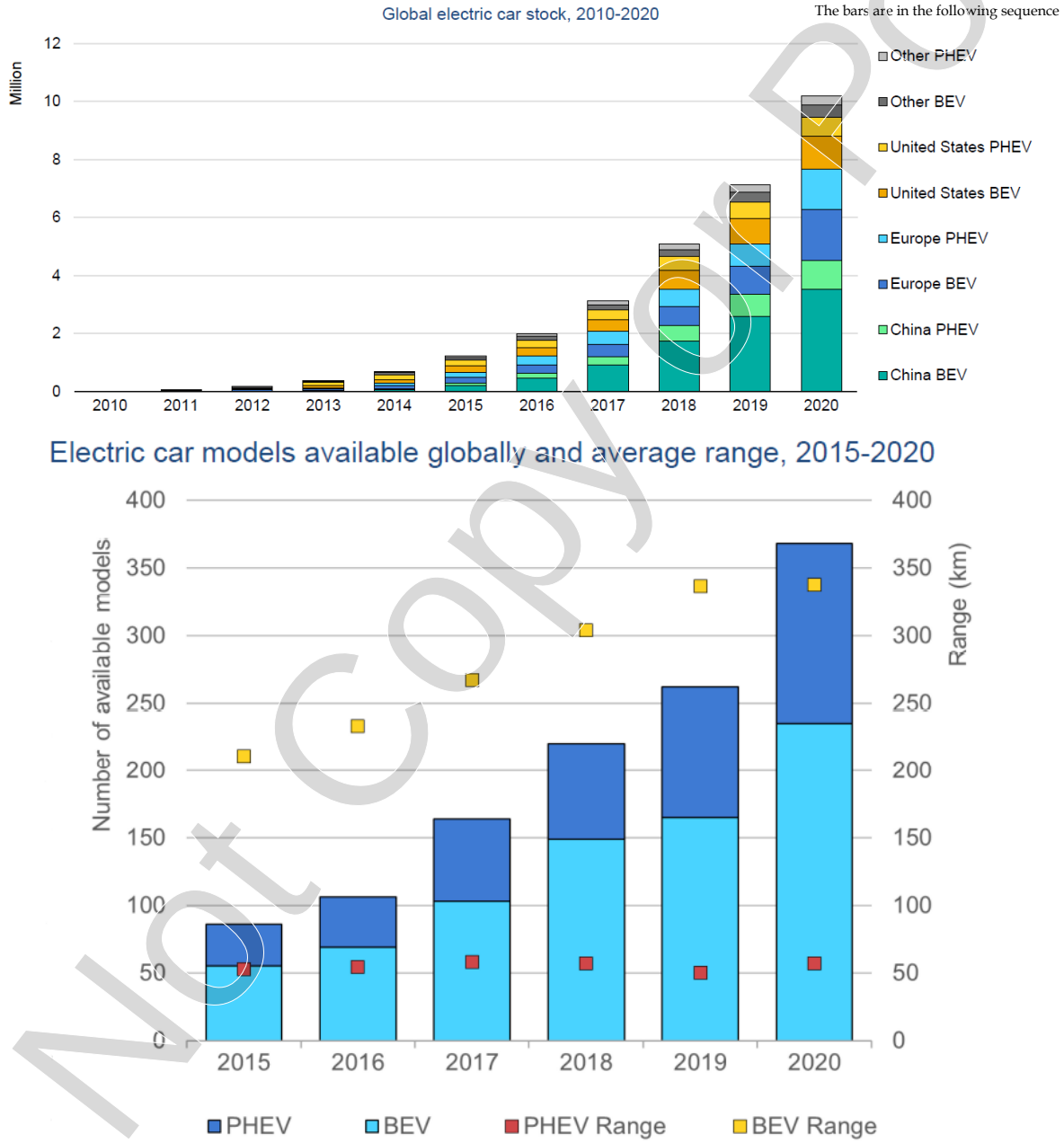
Source: The BMW Group, "The i3," https://www.bmw.co.uk/en/all-models/bmw-i/i3/2021/bmw-i3-highlights.html, "The i8," https://www.bmw.co.uk/en/topics/discover/bmw-i8.html, "The i4," https://www.bmw.co.uk/en/all-models/bmw-i/i4/2021/bmw-i4-highlights.html, all accessed July 2022.
Range and Mileage are from the U.S. Environmental Protection Agency (EPA), http://www.fueleconomy.gov.

Notes: MPGe refers to miles per gallon of gasoline-equivalent, a fuel efficiency metric for vehicles that rely on non-liquid fuels like electricity, hydrogen, and compressed natural gas. 60 miles per hour is equivalent to 96 kilometers per hour.

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Exhibit 4 Growth of the EV Market



Source: International Energy Agency (IEA), "Global EV Outlook 2021: Trends and Developments in Electric Vehicle Markets," 2021, https://www.iea.org/reports/global-ev-outlook-2021/trends-and-developments-in-electric-vehicle-markets, accessed August 2022.

Notes: BEV refers to battery electric vehicles. PHEV refers to plug-in hybrid electric vehicles.

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Exhibit 5 Charging Network Density in European Union Countries, as of 2020

Electric car charging points per 100km of road

By country, 2020

	Charging points per 100km
Netherlands	47.5
Luxembourg	34.5
Germany	19.4
Portugal	14.9
Austria	6.1
Belgium	5.5
Italy	5.1
Sweden	5.0
Denmark	4.4
France	4.1
Malta	3.4
Finland	3.3
Croatia	2.3
Slovakia	2.0
Slovenia	1.6
Spain	1.1
Ireland	1.0
Czech Republic	0.9
Bulgaria	0.8
Estonia	0.7
Hungary	0.6
Cyprus	0.5
Latvia	0.5
Romania	0.5
Poland	0.4
Greece	0.2
Lithuania	0.2

Source: European Automobile Manufacturers' Association (ACEA), "Electric Cars: 10 EU Countries do not have a Single Charging Point per 100km of Road," September 9, 2021, https://www.acea.auto/press-release/electric-cars-10-eu-countries-do-not-have-a-single-charging-point-per-100km-of-road/, accessed August 2022.

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Exhibit 6 Average Distance Driven per Day

Average Distance Driven per Day (in miles)

By country, 2017

	US	China	EU
Total	25.9	20.4	19.7
Urban	24.1	-	-
Suburban	24.8	-	-
Rural	33.0	-	-

Source: US: Transportation Energy Data Book, https://tedb.ornl.gov/wp-content/uploads/2022/03/TEDB_Ed_40.pdf; China: Statista, https://www.statista.com/statistics/1266966/china-average-annual-mileage-of-vehicles-in-use; EU: Odyssee-Mure, https://www.odyssee-mure.eu/publications/efficiency-by-sector/transport/distance-travelled-by-car.html. All accessed September 2022.

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Exhibit 7 BMW Financial Statements

Panel A: Income Statement data for BMW Group, 2017 to 2021

In € million	2021	2020	2019	2018	2017
Revenues	111,239	98,990	104,210	96.855	98,282
Cost of sales	(89,253)	(85,408)	(86,147)	(78,477)	(78,329)
Gross profit	**21,986**	**13,582**	**18,063**	**18,378**	**19,953**
Selling and administrative expenses	(9,233)	(8,795)	(9,367)	(9,568)	(9,560)
Other operating income	1,702	916	1,031	774	720
Other operating expenses	(1,055)	(873)	(2,316)	(651)	(1,214)
Profit / loss before financial result	**13,400**	**4,830**	**7,411**	**8,933**	**9,899**
Result from equity accounted investments	1,520	920	136	632	739
Interest and similar income	135	116	179	397	201
Interest and similar expenses	(165)	(458)	(499)	(386)	(412)
Other financial result	1,170	(186)	(109)	51	248
Financial result	**2,660**	**392**	**(293)**	**694**	**776**
Profit / loss before tax	**16,060**	**5,222**	**7,118**	**9,627**	**10,675**
Income taxes	(3,597)	(1,365)	(2,140)	(2,530)	(2,000)
Profit / loss from continuing operations	-	3,857	4,978	7,097	8,675
Profit / loss from discontinued operations	-	-	44	(33)	-
Net profit / loss	**12,463**	**3,857**	**5,022**	**7,064**	**8,675**
Attributable to minority interest	81	82	107	90	86
Attributable to shareholders BMW AG	12,382	3,775	4,915	6,974	8,589
Basic earnings per share of common stock in €	19	5.73	7.47	10.6	13.07
Basic earnings per share of preferred stock in €	18.79	5.75	7.49	10.62	13.09

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Panel B: Balance Sheet data for BMW Group, 2017 to 2021 (as of December 31 of each year)

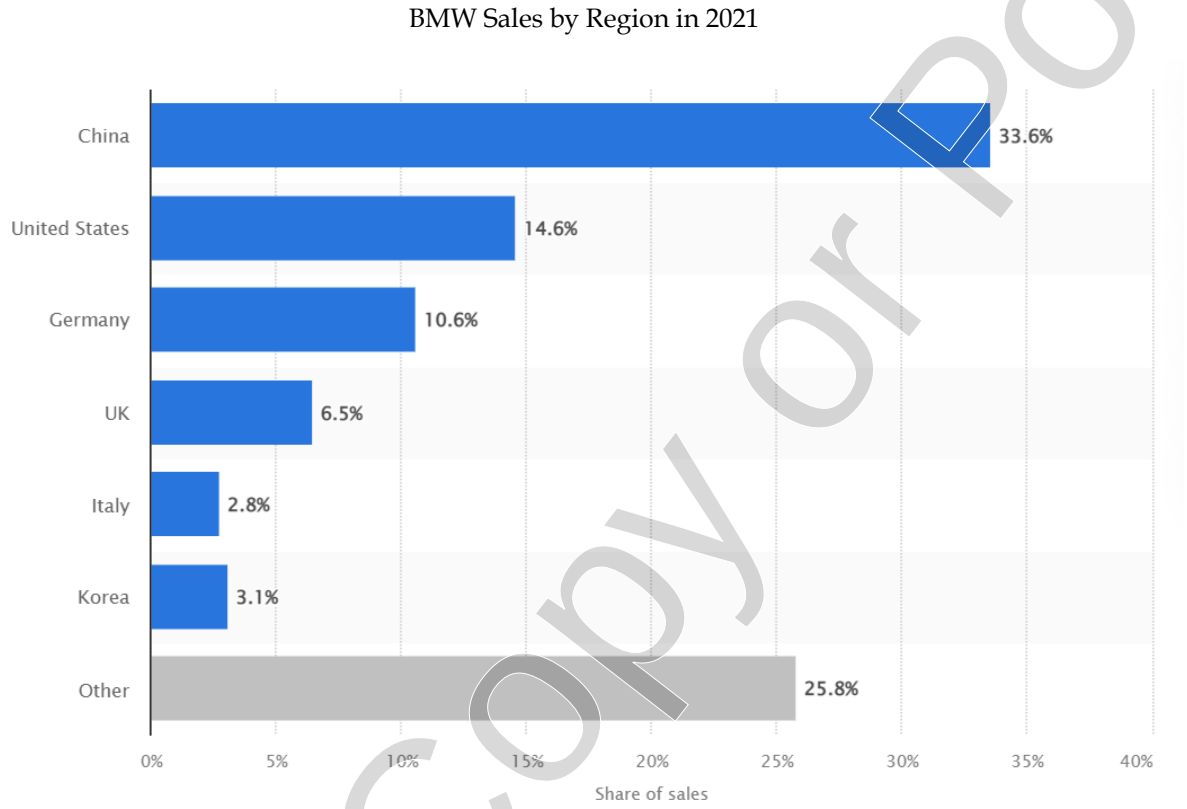
In € million	2021	2020	2019	2018	2017
ASSETS					
Intangible assets	12,980	12,342	11,729	10,971	9,464
Property, plant and equipment	22,390	21,850	23,245	19,801	18,471
Leased products	44,700	41,995	42,609	38,259	36,257
Investments accounted for using the equity method	5,112	3,585	3,199	2,624	2,769
Other investments	1,241	735	703	739	690
Receivables from sales financing	51,712	48,025	51,030	48,313	48,321
Financial assets	1,715	2,644	1,370	1,010	2,369
Deferred tax	2,202	2,459	2,194	1,638	1,993
Other assets	1,302	1,216	1,325	847	1,630
Non-current assets	**143,354**	**134,851**	**137,404**	**124,202**	**121,964**
Inventories	15,928	14,896	15,891	14,248	12,707
Trade receivables	2,261	2,298	2,518	2,546	2,667
Receivables from sales financing	35,705	36,252	41,407	38,700	32,113
Financial assets	5,800	5,108	5,955	6,675	7,965
Current tax	1,529	606	1,209	1,378	1,566
Other assets	8,941	9,110	11,614	9,749	7,485
Cash and cash equivalents	16,009	13,537	12,036	10,979	9,039
Assets held for sale	-	-	-	461	-
Current assets	**86,173**	**81,807**	**90,630**	**84,736**	**73,542**
Total assets	**229,527**	**216,658**	**228,034**	**208,938**	**195,506**
EQUITY AND LIABILITIES					
Subscribed capital	661	660	659	658	658
Capital reserves	2,325	2,199	2,161	2,118	2,084
Revenue reserves	71,705	59,550	57,667	55,862	50,815
Accumulated other equity	(325)	(1,518)	(1,163)	(1,338)	114
Equity attributable to shareholders of BMW AG	74,366	60,891	59,324	57,300	53,671
Minority interest	766	629	583	529	436
Equity	**75,132**	**61,520**	**59,907**	**57,829**	**54,107**
Pension provisions	1,247	3,693	3,335	2,330	3,252
Other provisions	7,206	6,488	5,788	5,530	5,632
Deferred tax	1,458	509	632	1,773	2,157
Financial liabilities	62,342	67,390	70,647	64,772	53,548
Other liabilities	5,676	5,095	5,100	5,293	5,045
Non-current provisions and liabilities	**77,929**	**83,175**	**85,502**	**79,698**	**69,634**
Other provisions	6,748	7,494	7,421	5,871	6,367
Current tax	921	747	963	1,158	1,124
Financial liabilities	41,121	38,986	46,093	38,825	41,100
Trade payables	10,932	8,644	10,182	9,669	9,731
Other liabilities	16,744	16,092	17,966	15,826	13,443
Liabilities in conjunction with assets held for sale	-	-	-	62	-
Current provisions and liabilities	**76,466**	**71,963**	**82,625**	**71,411**	**71,765**
Total equity and liabilities	**229,527**	**216,658**	**228,034**	**208,938**	**195,506**

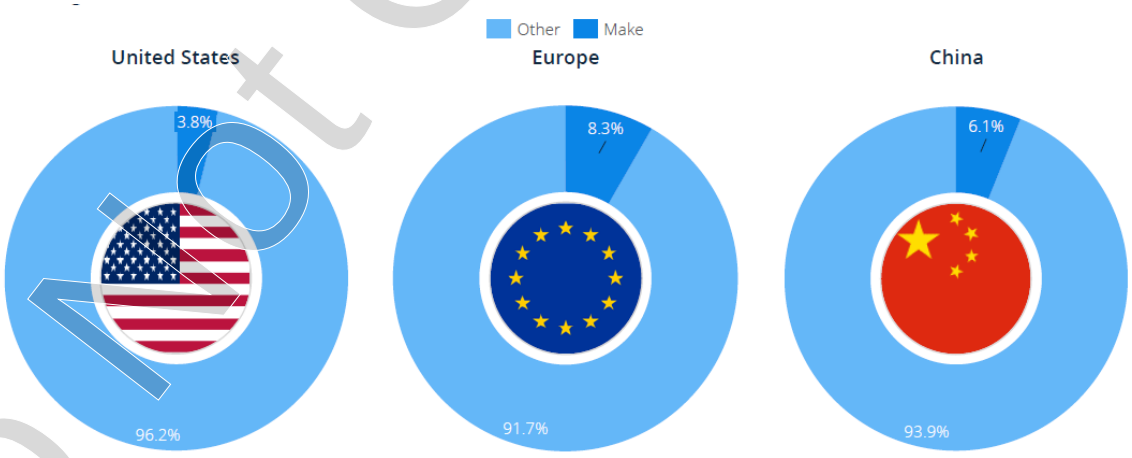
Source: BMW AG Financial Statements 2018 to 2021 spreadsheets, downloaded from https://www.bmwgroup.com/en/download-centre.html?fileType=spreadsheet, accessed July 2022.

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Exhibit 8 BMW Sales by Region





Source: Statista, https://www.statista.com/statistics/267252/key-automobile-markets-of-bmw-group/, accessed July 2022.

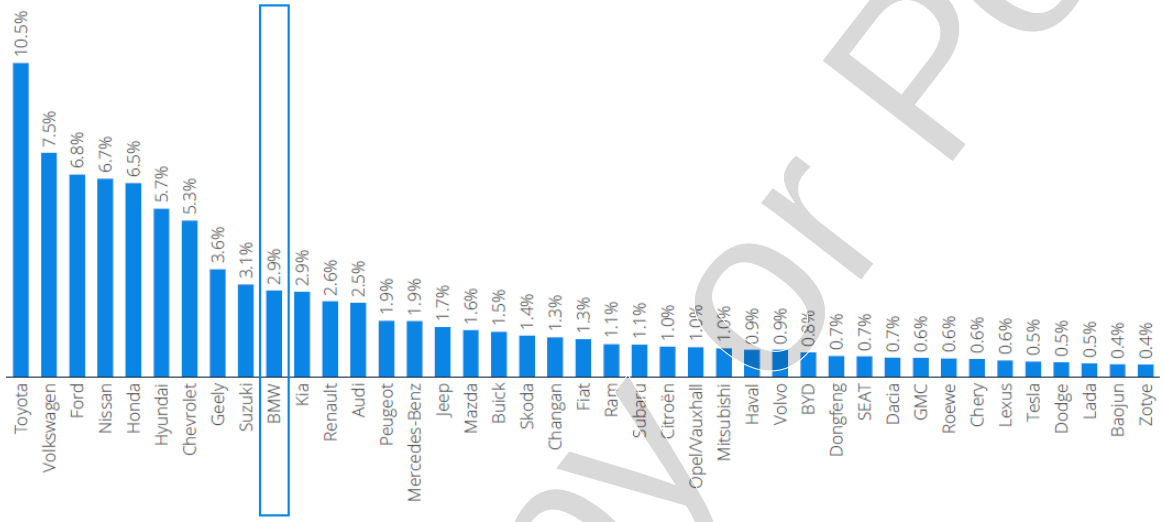
Notes: Make represents BMW sales

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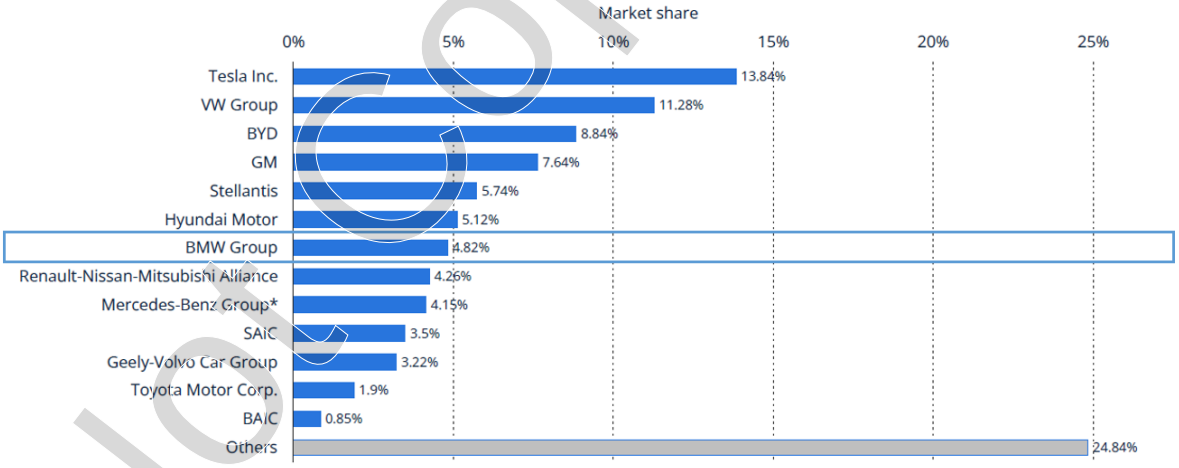
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Exhibit 9 Market Share

Market Share of Passenger Cars Unit Sales Worldwide in 2020



Market Share of Electric Vehicles Unit Sales Worldwide in 2021



Source: Statista, https://www.statista.com/study/60860/bmw-report/, accessed July 2022.

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Exhibit 10 Comparative Financial Metrics, end of year 2021

Company	Revenue (US$ billions)	P/E	P/B	Dividend yield	FCF yield	ROE	Earnings margin	Asset turnover	Leverage (A/E)
BMW	113	4.71	0.79	6.55	15.55	18.24	11.20	0.50	3.27
Mercedes-Benz	136	6.79	0.99	7.40	11.44	16.32	8.25	0.49	4.03
Volkswagen	255	5.99	0.62	4.26	18.11	11.22	6.17	0.49	3.73
Tesla	54	215.67	36.16	0.00	0.44	20.43	10.49	0.94	2.07

Source: Capital IQ, accessed August 2022.

Notes: Volkswagen brands also include SKODA, SEAT, CUPRA, Audi, Lamborghini, Bentley, Porsche, and Ducati.
All values are for fiscal year 2021 unless otherwise noted.
P/E = stock price as of the end of 2021 / earnings per share
P/B = stock price as of the end of 2021 / common equities as of the end of 2021
Dividend yield = dividends per share / stock price as of the end of 2021
FCF yield = free cash flow per share / stock price as of the end of 2021
ROE = Income from continued operations / average equities
Earnings margin = Income from continued operations / revenue
Asset turnover = Revenue / average assets
Leverage = Average assets / average equities

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Exhibit 11 Corporate-Wide Carbon Emissions of BMW Group, Mercedes-Benz, and Audi, 2019-2021

CO_2 Emissions by Scope and Brand, in million metric tons of CO_2e

	BMW			Mercedes-Benz			Audi		
	2021	**2020**	**2019**	**2021**	**2020**	**2019**	**2021**	**2020**	**2019**
Total emissions	122.54	118.49	133.55	124.29	NA	NA	NA	NA	NA
Scope 1	0.70	0.68	0.68	0.68	1.03	1.24	0.17	0.17	0.20
Scope 2	0.13	0.13	0.35	0.47	1.04	1.28	0.06	0.06	0.25
Scope 3	121.71	117.68	132.52	123.14	NA	NA	NA	NA	NA
Supply chain	18.53	16.23	18.51	20.40	NA	NA	NA	NA	NA
Use phase	99.81	98.78	110.90	99.00	NA	NA	NA	NA	NA
Other	3.37	2.67	3.11	3.74	NA	NA	NA	NA	NA
Deliveries (million vehicles)	2.52	2.33	2.54	2.33	2.46	2.82	1.69	1.70	1.85
Share of EVs (%)	13.0	8.3	5.8	9.8	7.4	2.0	NA	NA	NA
EU CO_2 emissions (in g / km)	115.9	99.1	127	115	104	137	122.1	102.9	130.6

Source: *BMW Group Report 2021*, https://www.bmwgroup.com/content/dam/grpw/websites/bmwgroup_com/ir/downloads/en/2022/bericht/BMW-Group-Report-2021-en.pdf, "Mercedes-Benz Group ESG Conference 2022 Presentation," https://group.mercedes-benz.com/dokumente/investoren/praesentationen/mercedes-benz-ir-esg-conference-2022-presentation.pdf, *Audi Report 2021*, https://www.audi.com/content/dam/gbp2/en/company/investor-relations/reports-and-key-figures/annual-reports/audi-report-2021.pdf, all accessed July 2022.

Notes: *Scope 3 – Other* for BMW refers primarily to vehicle disposal but also includes business trips and employee commuting.
Share of EVs refers to the proportion of the company's total deliveries in a given year that were EVs, based on vehicle volume.
EU CO_2 emissions refers to the average emissions per kilometer that result from driving the company's new vehicle fleet sold in EU that year. This average is weighted by the volume of vehicles sold across each of the company's models. The testing procedure was updated from the New European Driving Cycle (NEDC) to the Worldwide Harmonised Light Vehicles Test Procedure (WLTP) starting in 2021.

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Exhibit 12 Decarbonization Targets

Company	By	Targets
BMW	2025	• Increase share of BEVs in total deliveries to at least 30%
	2030	• Increase share of BEVs in total deliveries to at least 50% • Reduce CO_2 emissions per vehicle from production (Scope 1 + 2) by 80% (compared to 2019) (SBTi) • Reduce CO_2 emissions per vehicle in the use phase by 50% (compared to 2019) (SBTi) • Reduce CO_2 emissions per vehicle in the supply chain by 22% (compared to 2019) (SBTi)
	2050	• Achieve climate neutrality for total emissions
Mercedes-Benz	2025	• Increase share of PHEVs and BEVs in total deliveries to at least 50% • Establish a green steel supply chain
	2030	• Reduce absolute scope 1 and 2 GHG emissions 50% by 2030 (compared to 2018) (SBTi) • Reduce scope 3 GHG emissions from use of sold products 42% per vehicle kilometer (compared to 2018) (SBTi) • Reduce lifecycle CO_2 emissions per vehicle by 50% (compared to 2020) • Sell only BEVs wherever market conditions allow • Use at least 70% of energy generated by renewables (solar and wind power, via on-site installation and power purchase agreements [PPAs]) • Increase the use of recycled material per vehicle by 40%
	2039	• Achieve climate neutrality for total emissions
Audi	2025	• Increase share of PHEVs and BEVs in total deliveries to at least 40% • Reduce lifecycle CO_2 emissions per vehicle by 30% (compared to 2015)
	2030	• Reduce lifecycle CO_2 emissions per vehicle by 40% (compared to 2018)
	2033	• Gradual phase out of production of ICE vehicles
	2050	• Achieve climate neutrality for total emissions
Tesla	2030	• Increase annual sales of BEVs to 20 million per year (compared to 0.94 million in 2021)

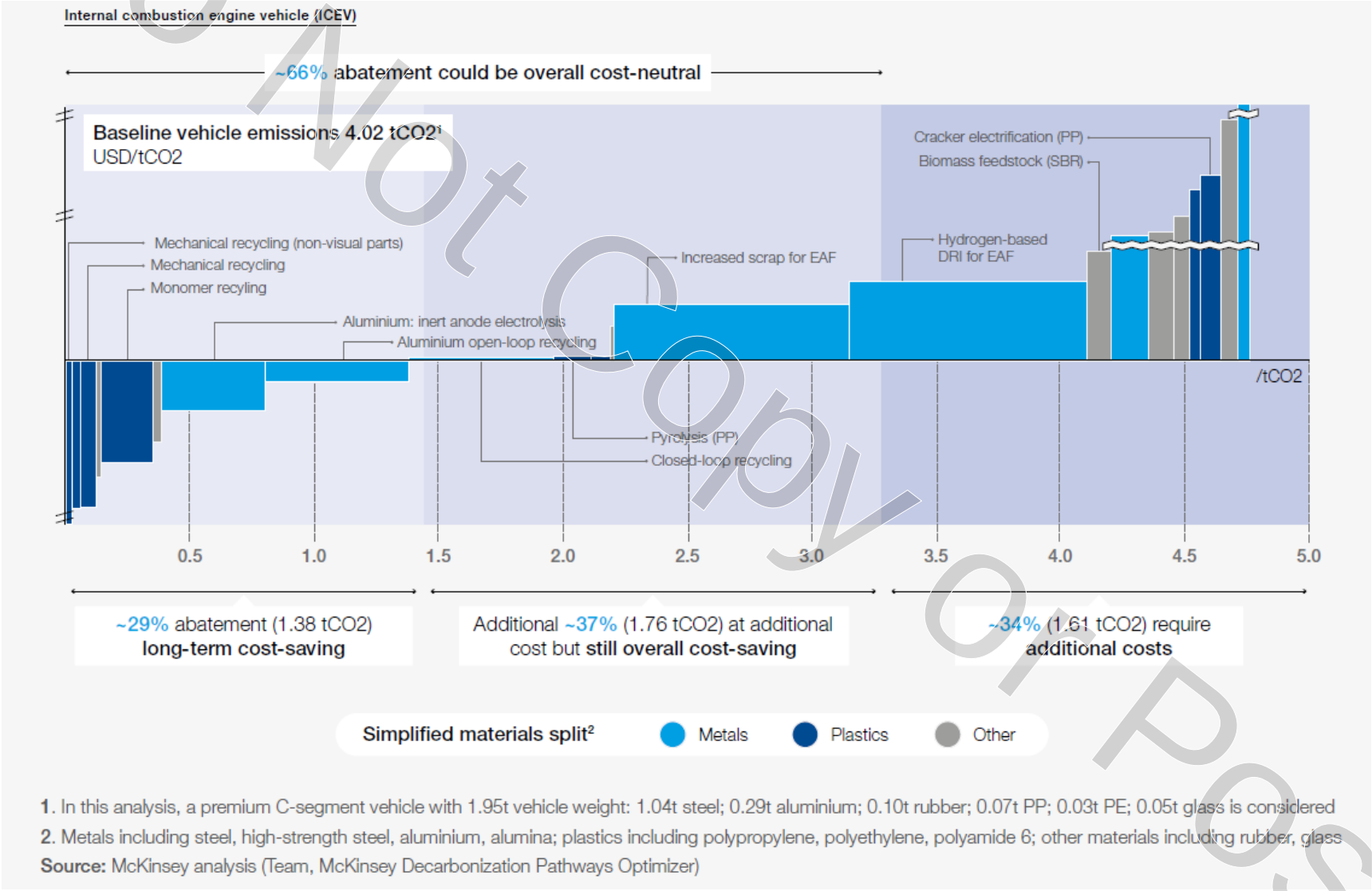
Source: *BMW Group Report 2021*, https://www.bmwgroup.com/content/dam/grpw/websites/bmwgroup_com/ir/downloads/en/2022/bericht/BMW-Group-Report-2021-en.pdf, "Mercedes-Benz Group ESG Conference 2022 Presentation," https://group.mercedes-benz.com/dokumente/investoren/praesentationen/mercedes-benz-ir-esg-conference-2022-presentation.pdf, *Audi Report 2021*, https://www.audi.com/content/dam/gbp2/en/company/investor-relations/reports-and-key-figures/annual-reports/audi-report-2021.pdf, *Tesla Impact Report 2021*, https://www.tesla.com/ns_videos/2021-tesla-impact-report.pdf, all accessed July 2022.

Notes: *SBTi* refers to targets approved by the Science-based Targets Initiative. Audi is a part of Volkswagen, and Volkswagen as a whole has targets validated by SBTi.

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Exhibit 13 Carbon Abatement Cost Curve



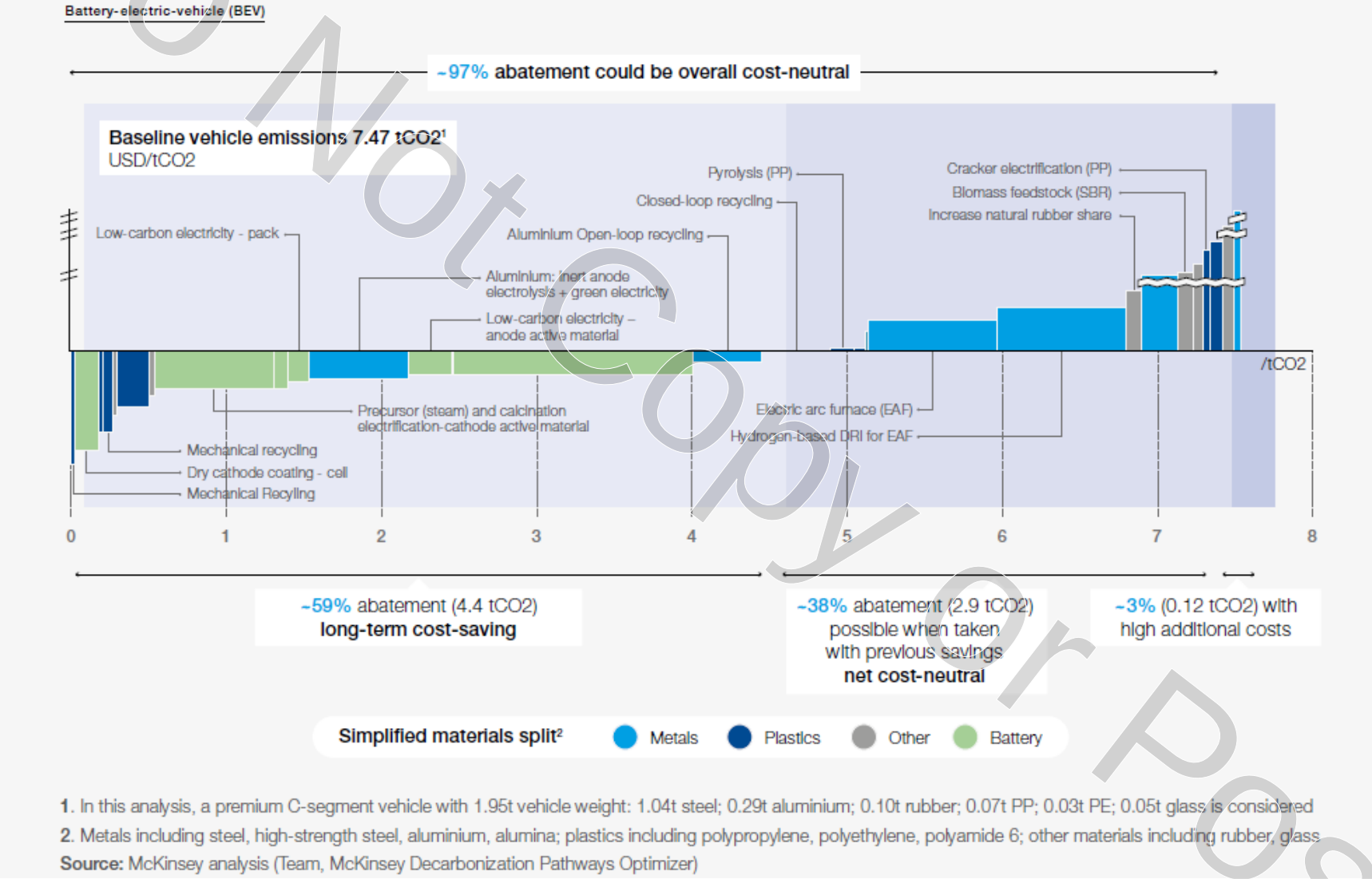
1. In this analysis, a premium C-segment vehicle with 1.95t vehicle weight: 1.04t steel; 0.29t aluminium; 0.10t rubber; 0.07t PP; 0.03t PE; 0.05t glass is considered
2. Metals including steel, high-strength steel, aluminium, alumina; plastics including polypropylene, polyethylene, polyamide 6; other materials including rubber, glass

Source: McKinsey analysis (Team, McKinsey Decarbonization Pathways Optimizer)

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Exhibit 13 (continued)



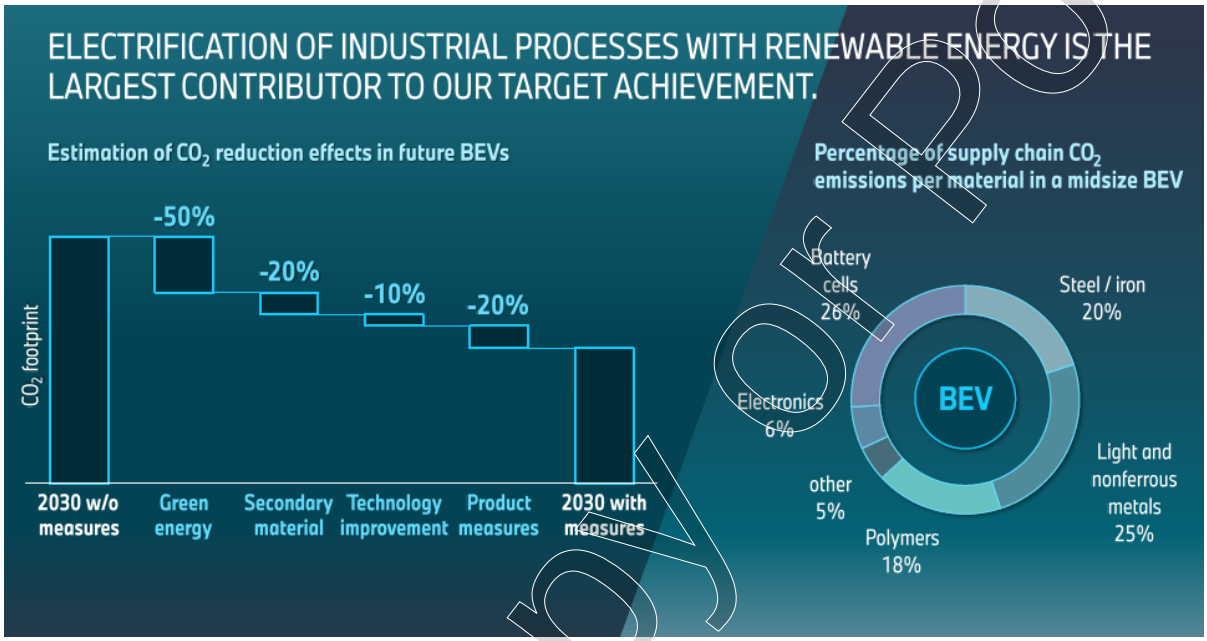
1. In this analysis, a premium C-segment vehicle with 1.95t vehicle weight: 1.04t steel; 0.29t aluminium; 0.10t rubber; 0.07t PP; 0.03t PE; 0.05t glass is considered
2. Metals including steel, high-strength steel, aluminium, alumina; plastics including polypropylene, polyethylene, polyamide 6; other materials including rubber, glass
Source: McKinsey analysis (Team, McKinsey Decarbonization Pathways Optimizer)

Source: World Economic Forum and McKinsey & Company, *Forging Ahead A Materials Roadmap for the Zero-carbon Car*, December 2020, https://www3.weforum.org/docs/WEF_Forging_Ahead_2020.pdf, accessed August 2022.

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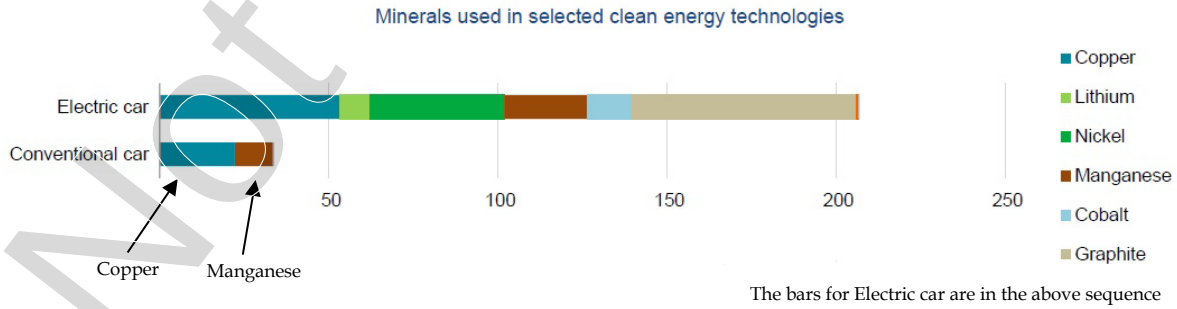
Exhibit 14a Materials Used in the Production of a BMW Battery Electric Vehicle



Source: Company materials.

Notes: The pie chart on the right shows the percent of supply chain CO_2 emissions by material used in the production of a BEV. The chart on the left shows BMW's pathway to achieving its 40% emissions reduction target by 2030.

Exhibit 14b Materials Used in Electric Vehicles and Internal Combustion Engine Vehicles (kg/vehicle)



Source: International Energy Agency (IEA), *The Role of Critical World Energy Outlook Special Report Minerals in Clean Energy Transitions* (Paris, 2021) https://www.iea.org/reports/the-role-of-critical-minerals-in-clean-energy-transitions, accessed August 2022.

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Exhibit 15 Carbon Intensity Emissions of Electricity Grids by Region, g CO_2 per kWh

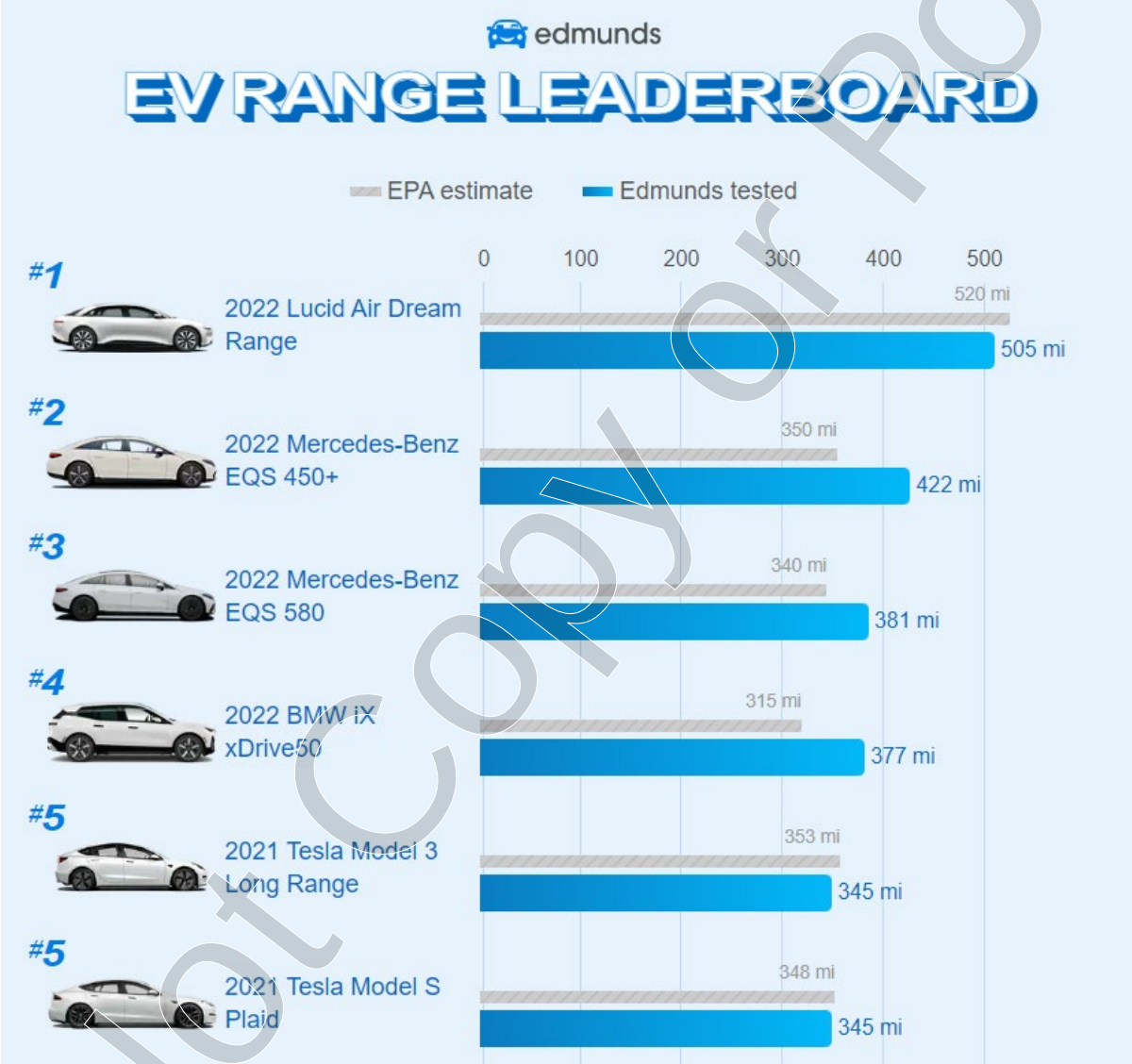
	2020	2019	2018	2017
China	611	623	636	630
U.S.	348	379	406	418
EU	215	239	273	290
Germany	311	338	406	424
U.K.	181	228	250	264

Source: China, US, EU: International Energy Agency, https://www.iea.org/reports/tracking-power-2021; Germany, UK: 2017-2019: European Environment Agency, https://www.eea.europa.eu/data-and-maps/indicators/overview-of-the-electricity-production-3/assessment; 2020 Germany: European Environment Agency, https://www.eea.europa.eu/ims/greenhouse-gas-emission-intensity-of-1; 2020 UK: National Grid ESO, https://www.nationalgrideso.com/news/record-breaking-2020-becomes-greenest-year-britains-electricity. All accessed July 2022.

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Exhibit 16 Battery Electric Vehicle Models with the Longest Range as of July 2022



Source: Jonathan Elfalan, "Edmunds Tested: Electric Car Range and Consumption - Real World vs. EPA," Edmunds, July 20, 2022, https://www.edmunds.com/car-news/electric-car-range-and-consumption-epa-vs-edmunds.html, accessed August 2022.

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